Homework 5 Template: Model-Based and Offline RL

CMU 10-703: Deep Reinforcement Learning (Fall 2026)

Use this template to record your answers for Homework 5. Add your answers using LaTeX and then save your document as a PDF to upload to Gradescope. You are required to use this template to submit your answers. **You should not alter this template in any way** other than to insert your solutions. You must submit all 13 pages of this template to Gradescope. Do not remove the instructions page(s). Altering this template or including your solutions outside of the provided boxes can result in your assignment being graded incorrectly.

You should also zip your `code/` directory and upload it to the **separate** Gradescope coding assignment. Remember to mark all teammates on **both** assignment uploads through Gradescope.

Instructions for Specific Problem Types

On this homework, you must fill in blanks for each problem. Please make sure your final answer is fully included in the given space. **Do not change the size of the box provided.** For short answer questions you should **not** include your work in your solution. Only provide an explanation or proof if specifically asked. To include a plot, use `\includegraphics[width=\linewidth]{plot.png}` inside the box, scaled so that it fits.

Fill in the blank: What is the course number? 10703

Problem 0: Collaborators

Enter your team members' names and Andrew IDs in the boxes below. If you worked in a team with fewer than three people, leave the extra boxes blank.

Name 1:		Andrew ID 1:	
Name 2:		Andrew ID 2:	
Name 3:		Andrew ID 3:	

Collaboration: Detail the work division amongst your group below.

Solution

Part I: Model-Based RL

Problem 1: Model-Based RL with PETS (105 pts)

1.1 Model-Based Predictive Control (25 pts)

1.1.1 (10 pts) CEM (without MPC) with ground-truth dynamics.

Success percentage:

1.1.2 (10 pts) Random sampling with ground-truth dynamics.

Success percentage without MPC:

Success percentage with MPC:

How does the performance of random sampling compare to that of CEM?

Solution

1.1.3 (5 pts) MPC vs. open-loop control: which performs better on what environments, and why? Discuss the pros and cons of MPC.

Solution

1.2 Single Probabilistic Network (30 pts)

1.2.1 (4 pts) Training loss plot.

Solution

1.2.2 (10 pts) MPC with random sampling. Success percentage:

1.2.3 (10 pts) MPC with CEM. Success percentage:

1.2.4 (6 pts) Random sampling vs. CEM. How did MPC with this model perform in general? When does the derived policy succeed, and when does it fail?

Solution

1.3 MBRL with PETS (15 pts)

1.3.1 (10 pts) Training loss plot.

Solution

1.3.2 (5 pts) Test success percentage vs. PETS iterations plot.

Solution

1.4 Training with Synthetic Data (35 pts)

1.4.1 (10 pts) Ablation over model vs. real data: plot and discussion.

Solution

1.4.2 (10 pts) Ablation over model rollout lengths: plot and discussion.

Solution

1.4.3 (10 pts) Model error vs. rollout length: plot and discussion.

Solution

1.4.4 (5 pts) Under which scenarios would you prefer MBRL to policy gradient methods?

Solution

Part II: Offline RL

Problem 2: TD3 + Behavior Regularization (50 pts)

2.2 Running TD3 without Any Offline Learning Modifications

Learning curves (evaluation returns and Q-values), and: what insights can we draw from evaluation returns being low but predicted Q-values being high?

Solution

2.3 Behavior Regularized TD3

Learning curves (evaluation returns and Q-values) with $\alpha_{\text{reg}} = 10$.

Solution

2.4 Conceptual Questions about Picking α_{reg}

1. For which kinds of datasets will we expect larger values of α_{reg} to work better? For which datasets will we expect lower values to work better?

   Solution

2. Given a completed run with $\alpha_{\text{reg}} = 1.0$, how would you choose a reasonable value of α_{reg} if you could only run a single more run?

Solution

Problem 3: Conceptual Questions on Offline RL (50 pts)

(i) Select all that apply and justify (see the handout for the three methods and statements 1–3).

Solution

(ii) Describe a scenario where distribution constraint methods would be preferable to support constraints.

Solution

(iii) Select all that apply and justify (see the handout for statements 1–3).

Solution

(iv) Why can offline RL perform better than imitation learning? Give a dataset where offline RL

should outperform imitation, and one where imitation should outperform offline RL.

Solution

Feedback

Feedback: You can help the course staff improve the course for future semesters by providing feedback. What was the most confusing part of this homework, and what would have made it less confusing?

Solution

Time Spent: How many hours did you spend working on this assignment? Your answer will not affect your grade. Please average your answer over all the members of your team.

Solution

Alone	
With teammates	
With other classmates	
At office hours	